

fraction expansion as $[1, 2, 2, 2, 2, \dots]$. This is the continued fraction expansion of $\sqrt{2}$, another irrational number that can't be represented as an integer fraction, has no pattern in its decimal digit, and yet has a convenient and easily memorable representation as a continued fraction.

From Fractions to Radicals

If you're interested in continued fractions, I recommend that you read about Srinivasa Ramanujan, who during his short life traveled mentally to the edges of infinity and brought some gems back for us to treasure. In addition to continued fractions, Ramanujan was interested in *continued square roots* (also known as *nested radicals*)—for example, the following three infinitely nested radicals:

$$x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$$

and

$$y = \sqrt{1 + 2 \times \sqrt{1 + 3 \times \sqrt{1 + 4 \times \sqrt{1 + \dots}}}}$$

and

$$z = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}$$

It turns out that $x = 2$ (an old anonymous result), $y = 3$ (as proved by Ramanujan), and z is none other than phi, the golden ratio! I encourage you to try to think of a method for generating nested radical representations in Python. Square roots are obviously interesting if we take them to infinite lengths, but it turns out that they're interesting even if we just consider them alone.

Square Roots

We take handheld calculators for granted, but when we think about what they can do, they're actually quite impressive. For example, you may